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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	P Tanvitha
Roll Number	2520030385
Branch	CSE
Course Outcome	CO-5

Case Study Topic: Cricket Scoring Service — Sort 1 Million Ball-by-Ball Delivery Records by (Over, Ball) Using Counting Sort (LSD Radix Style)

Work Done Summary:

Implemented counting sort twice (LSD radix-style) to sort cricket deliveries by composite key (over, ball). First pass sorts by ball-number (1..6) and the second stable pass sorts by over-number (0..49). Demonstrated bucket arrangement after each pass on the given 10-delivery example and explained the importance of stability for obtaining correct lexicographic order by (over, ball).

Implementation Details:

1. Input: 10 deliveries as (over, ball) pairs.
2. Pass 1 (LSD): Counting sort by ball ∈ [1..6] using 6 buckets; stable placement.
3. Pass 2: Counting sort by over ∈ [0..49] using required buckets; stable placement.
4. Output: Correct order by (over, ball).

Result: Screenshots / Output

```
PS C:\DSA-2> javac CricketCountingSort.java
PS C:\DSA-2> java CricketCountingSort

Input (unsorted) - 10 (over, ball) pairs
(2,4) (1,1) (3,6) (1,5) (2,2) (3,1) (1,3) (2,6) (3,4) (1,2)

Step 1 - Bucket by over (0..3 needed)
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over = 0 bucket :  []                        over = 2 bucket : (2,4) (2,2) (2,6)
over = 1 bucket :  (1,5) (1,1) (1,3) (1,2)   over = 3 bucket : (3,6) (3,1) (3,4)
over = 0 bucket :  []

Step 2 - Within each bucket, sort by ball (stable counting sort)
----------------------------------------------------------------
over = 0 bucket (sorted by ball) : []                        over = 2 bucket (sorted by ball) :  (2,2) (2,4) (2,6)
over = 1 bucket (sorted by ball) : (1,1) (1,2) (1,3) (1,5)   over = 3 bucket (sorted by ball) :  (3,1) (3,4) (3,6)

Final sorted order (by over, then ball):
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(1,1) (1,2) (1,3) (1,5) (2,2) (2,4) (2,6) (3,1) (3,4) (3,6)

Time Complexity: O(n + k1 + k2)  where k1 = 6 (balls), k2 = number of over-buckets used (≤ 50)
Space Complexity: O(n + k1 + k2)
```

Why Stability is Essential:

The first pass (by ball) groups balls correctly inside each over. Stability preserves the relative order of deliveries with the same ball when the second pass (by over) is applied. Without stability, deliveries within the same over could get reordered, breaking the correct lexicographic order by (over, ball).

GitHub Link: https://github.com/pondurutanvitha-stack/DSA-2

Student Signature	Faculty Signature